Q13 — Deeper projects

Deep Learning Systems 2026 — AI Internals Module

Framing

These projects investigate *how* large language models represent, store, and retrieve factual knowledge, and how that knowledge is converted into an answer. A useful mental model is the "knowledge neuron" / key–value memory view of feed-forward layers [1, 2], combined with causal localization methods such as causal tracing [3] and more recent critiques of the "localize-then-edit" assumption [6]. The projects below ask you to reproduce, stress-test, and extend these ideas. Each project lists concrete deliverables; all code, logs, and a short write-up (max 6 pages) are required.

1 Project A: Localizing factual associations with causal tracing

Goal. Reproduce and critically evaluate causal tracing [3] on an open-weight model (e.g. GPT-2 XL or a 7B-class LLaMA-family model).

Tasks.

1. Build a probe set of subject–relation–object factual statements. You may reuse CounterFact/zsRE-style items [3, 9].
2. Implement the three-run tracing protocol: clean run, corrupted-subject run, and corrupted-with-restoration run; record the *indirect effect* per (layer, token) site.
3. Produce causal-tracing heatmaps and identify the mid-layer MLP peak over subject tokens.
4. Critically test whether the localized site predicts where editing works, following **(author?)** [6]: compare tracing peaks against where edits actually succeed.

Deliverables.

- Code + reproducible config; heatmaps for ≥ 20 facts.
- A plot of edit-success vs. tracing-peak layer, with discussion of any mismatch.
- A 2-paragraph critique of what causal tracing does and does not establish.

2 Project B: Knowledge editing and its side effects

Goal. Implement and compare at least two editing methods and measure locality, generalization, and ripple effects.

Tasks.

1. Implement a rank-one / closed-form MLP edit (ROME) [3] and a mass-editing or hypernetwork baseline (MEMIT [4] or MEND [5]).
2. Evaluate on efficacy, paraphrase generalization, and *locality* (unrelated facts preserved).
3. Measure *ripple effects*: whether logically entailed facts update consistently [7].
4. Report failure modes (over-generalization, catastrophic interference under sequential edits).

Deliverables.

- Table comparing methods on efficacy / generalization / locality / ripple metrics.
- A sequential-editing stability curve (performance vs. number of edits).
- Discussion linking observed failures to the key–value memory view [1].

3 Project C: Probing representations and the geometry of facts

Goal. Investigate how knowledge is encoded in hidden states and whether it is linearly accessible.

Tasks.

1. Train linear probes to recover attributes (e.g. entity type, country) from hidden states across layers; report accuracy by layer.
2. Use the logit-lens / early-decoding idea to track when the answer token becomes dominant across layers [2, 10].
3. Test the "linear representation" hypothesis for selected relations [8], estimating relation operators from hidden states.
4. Distinguish genuine knowledge from surface heuristics using controlled minimal pairs.

Deliverables.

- Layer-wise probe accuracy plots and logit-lens trajectories for ≥ 10 facts.
- A short analysis of whether the studied relations are approximately linear.
- Caveats on probing (correlation vs. causation; probe expressivity).

4 Project D: Parametric vs. retrieved knowledge

Goal. Compare answering from parametric memory against retrieval-augmented generation (RAG), and study conflict resolution.

Tasks.

1. Build a small RAG pipeline (dense retriever + reader) [11, 12].
2. Measure closed-book vs. open-book accuracy on a factual QA set [13].
3. Create knowledge-conflict cases (retrieved passage contradicts parametric belief) and quantify when the model defers to context [14].
4. Analyze hallucination: do parametric gaps correlate with confident wrong answers?

Deliverables.

- Accuracy comparison (closed vs. open book) with confidence calibration plots.
- A conflict-resolution table (defer-to-context rate by passage type).
- Recommendations on when retrieval should override parametric knowledge.

Assessment

Projects are graded on reproducibility (30%), correctness of implementation (30%), depth of analysis and critique (30%), and clarity of the write-up (10%). Negative or null results are fully acceptable when rigorously argued.

References

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